

ONTOLOGICAL PROTOCOL: TECHNICAL ADDENDUM v1.0

Critical Specification Fixes for Simulation Readiness

The Resonance Collective

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Abstract

This document serves as a mandatory technical addendum to the *Canonical Edition (CE)*. It addresses five specific implementation ambiguities identified during the architectural audit. These definitions take precedence over the simplified descriptions in the core CE text to ensure deterministic consensus and simulation stability.

1 Patch 1: The Cold Start (Genesis Provision)

Issue: The CE requires token burn $b(e)$ for valid events. New agents with zero balance cannot enter (Deadlock). **Fix:** We introduce *Thermodynamic Substitution*.

Spec-Patch 1 (Effective Burn). For any event e , the system calculates an *Effective Burn* $b_{eff}(e)$ used only for:

- Validity checking (minimum thermodynamic floor),
- Epoch budget computation.

It does **not** replace literal burn $b(e)$ inside Ripple recursion.

$$b_{eff}(e) = b(e) + \min(C_{cap}, \gamma_{pow} \cdot \text{Work}(e)) \quad (1)$$

Ripple anchor remains strictly:

$$R_T(e) \supset b(e)$$

Thus Proof-of-Work allows entry without granting structural mass.

Where:

- $b(e)$ is the literal token burn (can be 0).
- $\text{Work}(e)$ is the verified difficulty of the attached Proof-of-Work nonce.
- C_{cap} is a protocol constant limiting the maximum substitute value to prevent inflation via compute-spam.

Result: An agent can enter permissionlessly via energy expenditure without requiring pre-existing tokens.

2 Patch 2: Deterministic Neighbor Sampling

Issue: "Truncated neighbors" was underspecified. Nondeterministic truncation breaks consensus. **Fix:** Lexicographical sorting.

Spec-Patch 2 (Deterministic Truncation). Let $Children_H(e)$ be the set of events y such that:

$$e \in Parents(y) \quad \text{and} \quad T - H + 1 \leq T(y) \leq T$$

Define first-degree signer neighborhood:

$$N_1(e) = \bigcup_{y \in Children_H(e)} Sig(y) \setminus Sig(e)$$

Define second-degree neighborhood:

$$N_2(e) = \bigcup_{z \in Children_H(y), y \in Children_H(e)} Sig(z) \setminus (N_1(e) \cup Sig(e))$$

To compute the truncated set $\tilde{N}_1(e)$ of size K : 1. Sort the keys in $N_1(e)$ lexicographically (e.g., ascending hex value). 2. Select the first $\min(|N_1(e)|, K)$ keys. This ensures every node calculates the exact same topological entropy $\Omega(e)$.

3 Patch 3: Edge-Specific Persistence

Issue: Applying $\lambda_T(e)$ to the parent scales all children identically, ignoring the age of the specific link. **Fix:** Shift persistence to the edge.

Spec-Patch 3 (Edge Decay). In the Ripple recursion, replace $\lambda_T(e)$ with $\lambda_T(e, y)$:

$$\lambda_T(e, y) = \lambda_{base}(e) \cdot e^{-\frac{T(y) - T(e)}{\tau}} \quad (2)$$

Where $T(y)$ is the child's epoch and $T(e)$ is the parent's epoch. *Result:* A parent event keeps its structural depth (λ_{base}), but its influence on specific children decays over time unless renewed. This is deterministic truncation, not probabilistic sampling. All nodes apply identical ordering rules to ensure consensus.

4 Patch 4: Stasis Aggregation Order

Issue: If a signer executes multiple events in one epoch, the order of Stasis realization is ambiguous. **Fix:** Epoch-level Aggregation.

Spec-Patch 4 (Epoch Aggregation). Stasis is not calculated per event, but per epoch.

$$S_{real}(a, T) = \min \left(S_{pot}(a, T), \eta \sum_{e \in \mathcal{E}_T, a \in Sig(e)} DCD_T(e) \right) \quad (3)$$

The potential S_{pot} is deducted *after* the full epoch is processed. This removes dependency on intra-epoch sorting.

5 Patch 5: The Missing Regulator (Part V)

Issue: The CE references a regulator ϵ_T but does not define its update rule. **Fix:** Damped PID Logic.

Spec-Patch 5 (Homeostasis). Let σ_T be the measured global branching ratio in epoch T .

Define exponential smoothing:

$$\hat{\sigma}_T = (1 - \beta)\hat{\sigma}_{T-1} + \beta\sigma_T \quad (4)$$

where $0 < \beta \leq 1$.

Let $\sigma_{target} \approx 1.0$ be the target branching ratio.

The inflation scalar updates via proportional control:

$$\epsilon_{T+1} = \text{clip}_{[\epsilon_{min}, \epsilon_{max}]}(\epsilon_T + \kappa_P(\sigma_{target} - \hat{\sigma}_T)) \quad (5)$$

- If $\hat{\sigma} < 1$ (Stagnation): $\epsilon \uparrow$ (Minting increases, lowering relative cost of Burn).
- If $\hat{\sigma} > 1$ (Overheating): $\epsilon \downarrow$ (Minting decreases, raising relative cost of Burn).

Definition 1 (Global Branching Ratio).

$$\sigma_T = \frac{\sum_{e \in \mathcal{E}_T} |Children_H(e)|}{|\mathcal{E}_T|}$$